Contents

1 FPGA Peripherals **1**

2 Memory **2**
2.1 Types of memory 2
2.2 Memory Hierarchy Design 3

3 RAM **4**
3.1 SRAM 4
3.1.1 Large SRAM implementation 5
3.2 DRAM 5
3.2.1 DRAM vs SRAM 6
3.2.2 Asynchronous & Synchronous DRAM 6
3.2.3 Interleaving 7
3.2.4 Organization of the DRAM 7
3.2.5 DRAM Subsystem 9
3.2.6 Basic DRAM Controller Operation 10
3.2.7 Internal Physical Structure of DRAM 12
3.3 DDR RAM 12
3.4 DDR2 RAM 13
3.5 DDR3 RAM 14
3.6 DDR4 RAM 14
3.7 Application specific DDR versions 15
3.8 DDR Packaging 15
3.9 Xilinx DDR MIG Controller IP 15

4 Flash Memory **15**
4.1 QSPI Flash 16

5 EMMC **16**
5.1 eMMC Device Overview 17

6 Gigabit Ethernet **19**
6.1 1G/2.5G Ethernet IP 21
6.2 10G/25G Ethernet IP 23

7 Direct Memory Access **23**

1 FPGA Peripherals

- Memories
- RAM
- DDR
- DDR3
- Flash
- EMMC
- Gigabit Ethernet
- DMA

2 Memory

A memory unit is a device to which binary information is transferred for storage and from which information is retrieved when needed for processing. A memory unit is an integral part of any computing system, and its primary purpose is to hold instructions and data. When data processing takes place, information from memory is transferred to selected registers in the processing unit. Intermediate and final results obtained in the processing unit are transferred back to be stored in memory.

shows a logical picture of components of a Modern Computer. One can observe how different types of memories are interfaced with a processor.

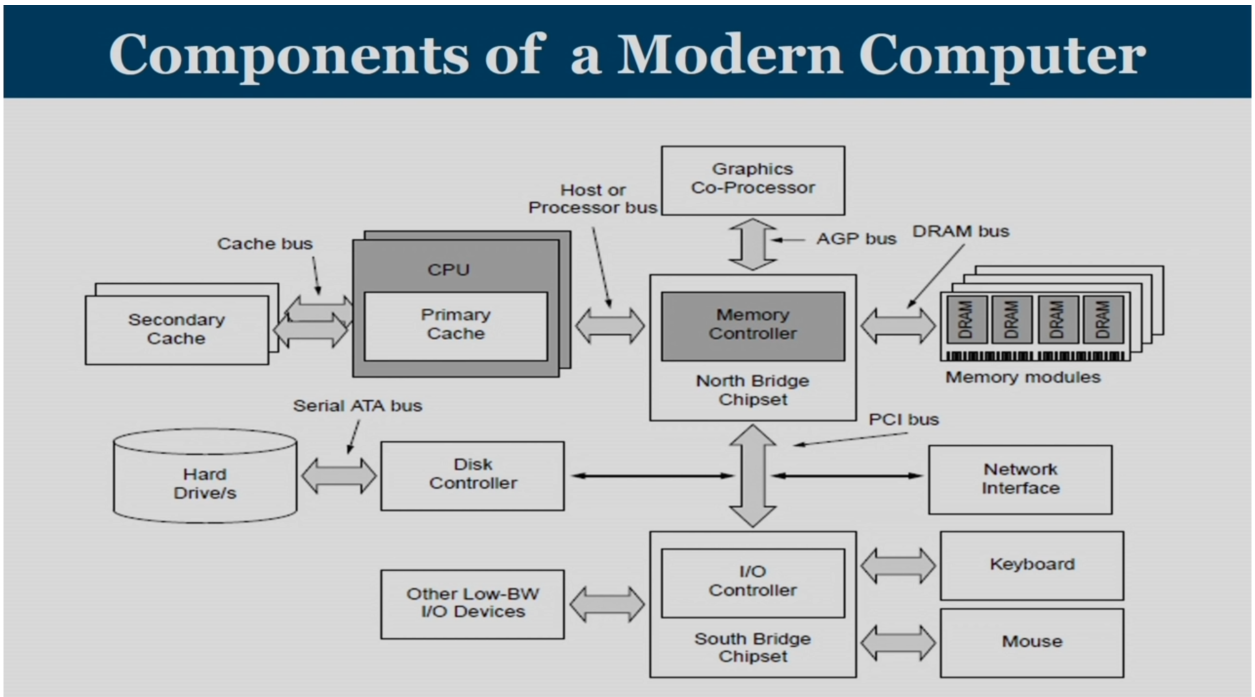

Figure 1: Component of a Modern Computer

2.1 Types of memory

In a broad sense, a memory system in any digital systems can be divided into two types:

1. Read Only Memory (ROM), and
2. Random Access Memory (RAM)

Read Only Memory (ROM) is a type of memory where the data has been prerecorded. This means that suitable binary information is already stored inside memory and can be retrieved or read at any time. However, that information cannot be altered by writing. Data stored in ROM is retained even after the computer is turned off non-volatile. There are four types of ROM:

Programmable ROM (PROM) where the data is written after the memory chip has been created. It is non-volatile.

Erasable Programmable ROM (EPROM) where the data on this non-volatile memory chip can be erased by exposing it to high-intensity UV light.

Electrically Erasable Programmable ROM (EEPROM) where the data on this non-volatile memory chip can be electrically erased using field electron emission.

Mask ROM in which the data is written during the manufacturing of the memory chip.

Random Access Memory (RAM) is used to store the programs and data being used by the CPU in real-time. The data on the random access memory can be read, written, and erased any number of times. RAM is a hardware element where the data being currently used is stored. It is a volatile memory which means that the data stored on the RAM gets erased on a power reset.

2.2 Memory Hierarchy Design

In the Computer System Design, Memory Hierarchy is an enhancement to organize the memory such that it can minimize the access time. The Memory Hierarchy was developed based on a program behavior known as locality of references. clearly demonstrates the different levels of memory hierarchy:

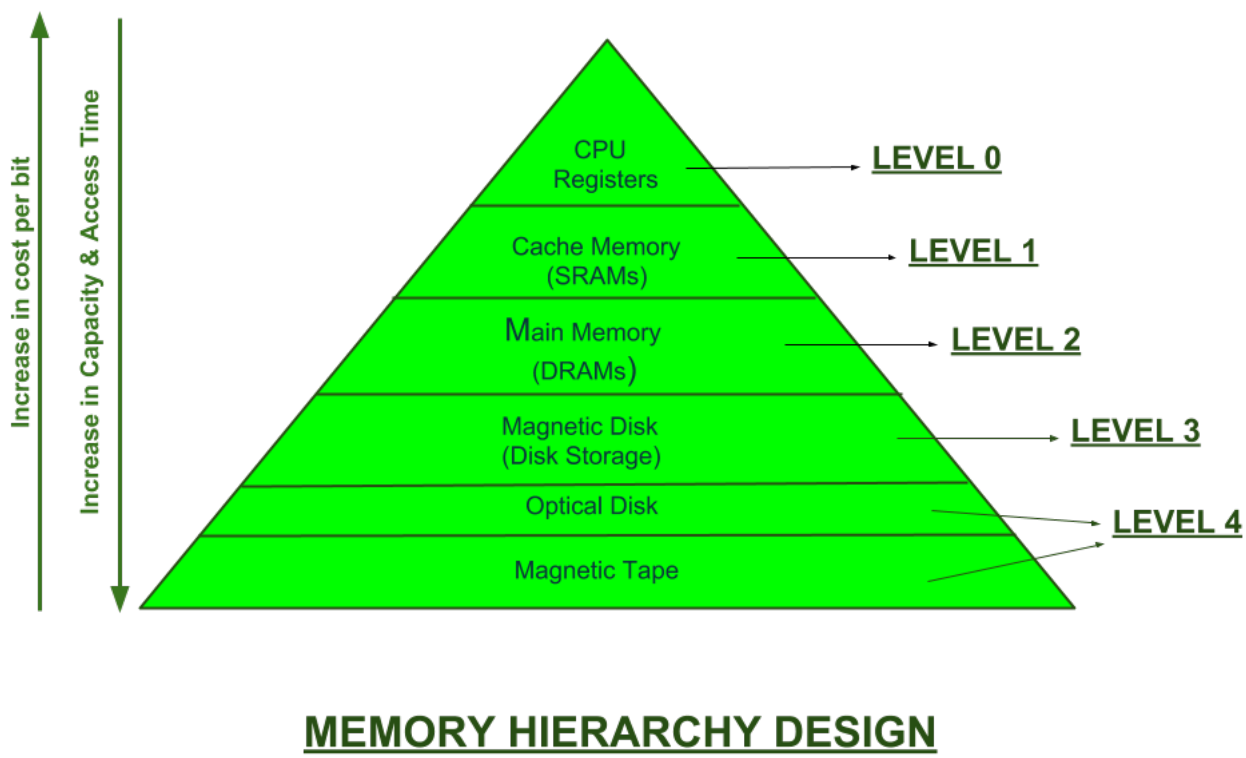

Figure 2: Component of a Modern Computer

This Memory Hierarchy Design is divided into 2 main types:

External Memory or Secondary Memory Comprising of Magnetic Disk, Optical Disk, Magnetic Tape peripheral storage devices which are accessible by the processor via I/O Module.

Internal Memory or Primary Memory Comprising of Main Memory, Cache Memory & CPU registers. This is directly accessible by the processor.

In the Internal Memory or Primary Memory, the entire program and data of a given application cannot be located fully inside local memories BRAMs, UltraRAMs(FPGAs), cache memories (processors). The memory is implemented as a hierarchy, where we have the registers inside the processor/FPGAs as the fastest memory, then we have caches/BRAMs. And, then the next level of memory is known as the main memory or DRAM system.

3 RAM

Random Access Memory (RAM) is the internal memory of the digital system for storing data, program, and program result. It is a read/write memory which stores data until the machine is working. There are two types of RAM:

Static RAM (SRAM) which stores a bit of data using the state of a six transistor memory cell.

Dynamic RAM (DRAM) which stores a bit data using a pair of transistor and capacitor which constitute a DRAM memory cell.

3.1 SRAM

Static random access memory (static RAM or SRAM) is a type of RAM that uses latching circuitry (flip-flop) to store each bit. shows an SRAM cell.

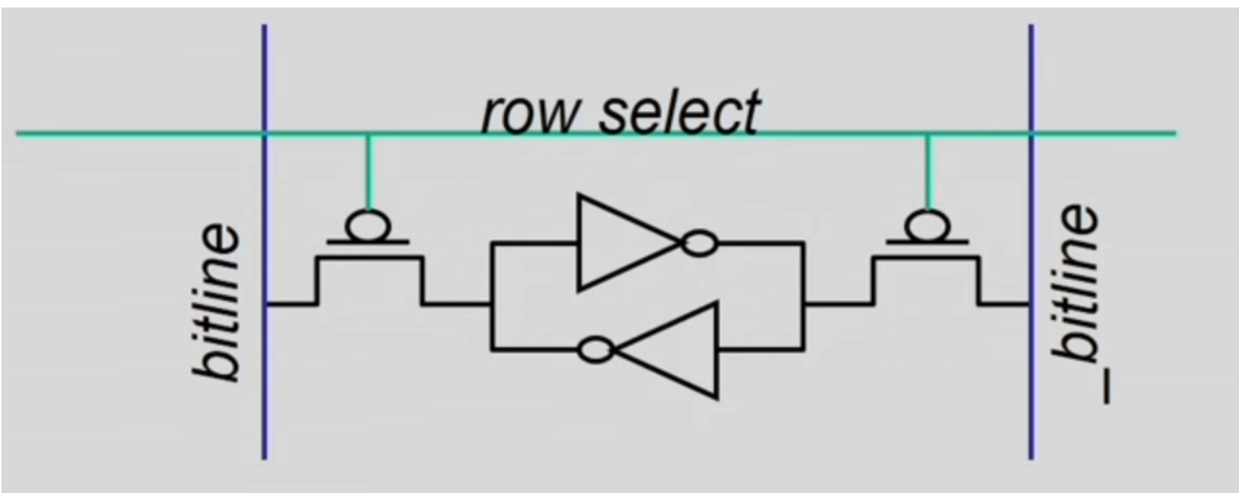

Figure 3: Static RAM cell

SRAM cell can store 1 bit of information which consists of a row line and a bitline. A pair of bit lines is used for storage of every bit, one is bitline and the other one is bitline compliment. Bitline compliment is a logical compliment of the bitline.

Two cross coupled NOT gates are connected by two transistor, which are connected to the row select. So, once a particular row is selected both T1 and T2 is going to be in on position. So, whatever value is there in the bitline it flows into the not gates. The value in the bitline will get stored inside the two cross coupled NOT gates loop. Hence, transistor will basically act as a switch in this context.

3.1.1 Large SRAM implementation

These SRAM memory cells, consisting of six transistors, organized as rows and columns to get an organized structure for the main memory. Address needs to be generated consisting two components, ‘n’ bits representing the rows and ‘m’ bits representing the columns.

While reading from the memory, the row is chosen by using an n to $2^{(}n)$ decoder.Once the row is selected, the entire contents of the row are transferred to a sense amplifier. And, the single bit extracted by selecting from ‘m’ column number.

Read sequence is as follows:

1. Address decode
2. Drive row select
3. Selected bit-cells drive bitlines (entire row is read together)
4. Column select (Data is ready)

3.2 DRAM

Dynamic random access memory (Dynamic RAM or DRAM) is a type of random access memory that stores each bit of data in a memory cell, usually consisting of a tiny capacitor and a transistor, both typically based on metal-oxide-semiconductor (MOS) technology.

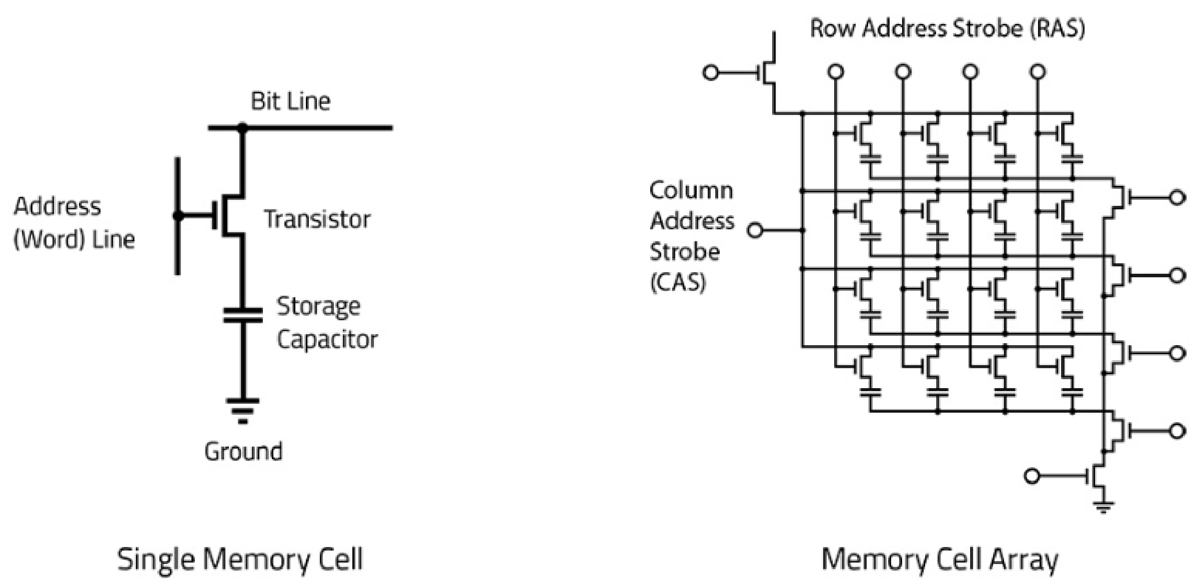

Figure 4: Dynamic RAM cell

Bits are basically stored as charges on the capacitor and a memory cell lose charge when it is read. When there exists a potential difference between the parallel plates of a capacitor, it is called as logic 1. And, when the potential difference between the two parallel plates of a capacitor is less than a threshold value, then it is called as logic 0.

- Bits are stored as charges on capacitor.
- Memory cell loses charge when read.
- Memory cell loses charge over time.
- A flip flop in sense amplifier amplifies and regenerates the bitline and data bit is multiplexed out of it.

Since the capacitor discharges over time, the information stored eventually fades unless the capacitor is periodically REFRESHed. This is where the 'D' in DRAM comes from. It refers to Dynamic as opposed to static in SRAM.

3.2.1 DRAM vs SRAM

DRAM

- Slower access (capacitor)
- Higher density (transistor, capacitor cell)
- Lower cost
- Requires refresh (power, performance, circuitry)
- Manufacturing requires putting capacitor and logic together

SRAM

- Faster access (no capacitor)
- Lower density (6 transistor cell)
- Higher cost
- No need for refresh
- Manufacturing compatible with logic process (no capacitor)

Density plays a crucial role in accommodating larger memory in a smaller size memory. DRAM is preferred in order to implement the primary memory.

3.2.2 Asynchronous & Synchronous DRAM

In different generations of dynamic RAM there is an improvement in the speed as well as the reduction in the power consumption. Asynchronous Dynamic RAM can be considered as the oldest generation of dynamic RAM. Asynchronous DRAM means that the RAM is not synchronized with the CPU clock. The obvious disadvantage of this particular type of RAM was that then CPU does not know the exact timing at which the data will be available from the RAM on the input output bus.

This problem has been overcome by the next generation of RAM, which is known as the synchronous DRAM. In case of SDRAM, the RAM is synchronized with the CPU clock. Now, the advantage of this type of SDRAM is that the CPU or to be precise, the DRAM memory controller knows the exact timing or the number of cycles after which the data will be available on the bus. And hence, the CPU does not need to wait for the memory access. This also results in increasing memory read & write speeds. The synchronous DRAM modules are operated at 3.3V. SDRAM or synchronous

DRAM is also known as the Single Data Rate SDRAM as the data is transferred at the every rising edge of the clock cycle.

3.2.3 Interleaving

One of the main issues in accessing a single monolithic memory is that A single monolithic memory array takes long to access and does not enable multiple accesses in parallel.

The solution to this problem is to divide the entire memory into multiple banks that can be accessed independently (in the same cycle or in consecutive cycles).

The key design issue is to map the data into different banks. This issue is resolved by the process referred to as Interleaving, or Banking. In interleaving,

- Address space partitioned into separate banks
- No increase in data store area
- Bits in address determines which bank an address maps to
- Cannot satisfy multiple accesses to the same bank
- Crossbar interconnect in input as well as output
- Bank conflicts: Two accesses to the same bank are difficult to handle

One simple way of implementing banking is odd even separation. All the even addresses can be considered as mapped to bank 0 and all the odd addresses can be considered as mapped to bank 1.

The address provided to read the data is typically referred as ”logical address”. Logical address is translated to a physical address before it is presented to the DRAM. The physical address is made up of the following fields:

- Bank Group
- Bank
- Row
- Column

These individual fields are then used to identify the exact location in the memory to read-from or write-to.

3.2.4 Organization of the DRAM

DRAM consists of multiple hierarchies of channels, DIMM(Dual Inline Memory Module), rank, chip, bank, row columns, and B-cells/ Memory Cells. A digital system can request data reads from memory at any point of time. To read from the memory, address has to be provided and to write to the memory, data & address has to be provided.

shows the organization of the DRAM.

- The processor may have multiple channels it may have multiple address buses or data buses.
- A channel is formed by joining multiple DIMMs.
- The DIMM has a front side (known as rank 0) and a back side (known as rank 1).

- Rank is a set of chips that respond to the same command and same address at the same time, but with different pieces of requested data.
- Both ranks are usually provided with a common address, and one rank, either rank 0 or rank 1, gives the corresponding data.
- Rank comprises of multiple chips.
- Each chip holds and shares a sub component of the entire data held by a rank.
- Each chip has a 3D structure consisting multiple banks with each bank consisting a layer of rows and columns.
- Breaking down a bank, each bank consists of rows as well as columns.

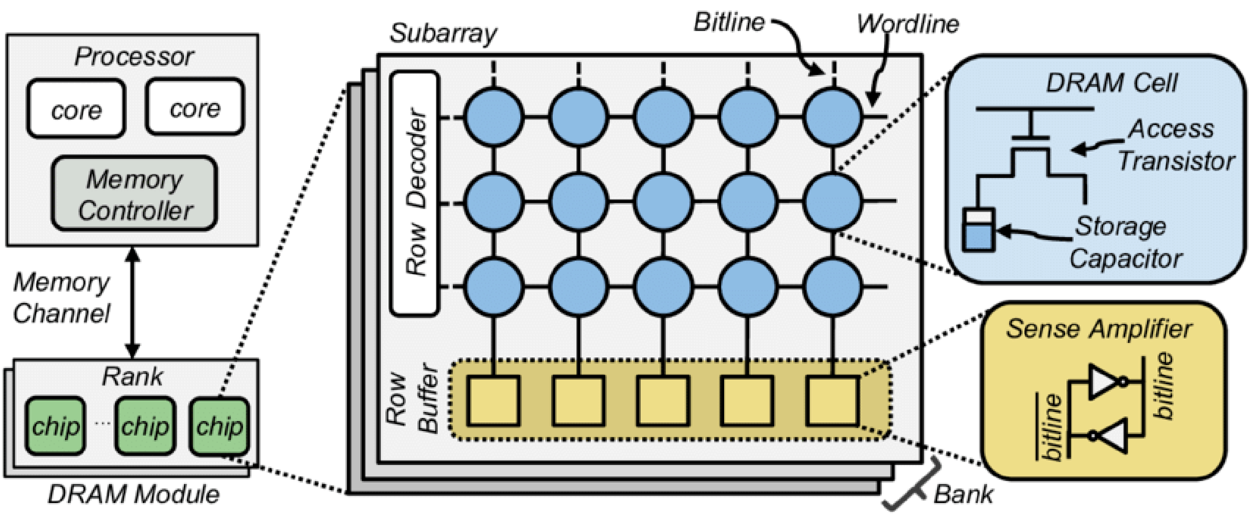

Figure 5: Organization of the DRAM

Going down another level, DRAM consists of a page mode structure. DRAM bank is a 2D array of cells which consists of rows and columns. Each Bank contains the following:

- Memory Arrays
- Row Decoder
- Column Decoder
- Sense Amplifiers

Once the Bank Group and Bank have been identified, the Row part of the address activates a line in the memory array. This is called the "Word Line" and activating it reads data from the memory array into "Sense Amplifiers". Sense amplifier are kept in row buffers. The Column address then reads out a part of the word that was loaded into the Sense Amplifiers. The width of the column is called the "Bit Line".

The width of a column is standard it is either 4 bits, 8 bits or 16 bits wide and DRAMs are classified as x4, x8 or x16 based on this column width. Another thing to note is that, the width of the data bus is same as the column width.

3.2.5 DRAM Subsystem

The DRAM talks to the ASIC or FPGA through the system called as the DRAM Subsystem. DRAM Subsystem is made up of 3 components:

- The DRAM memory
- A DRAM PHY
- A DRAM Controller

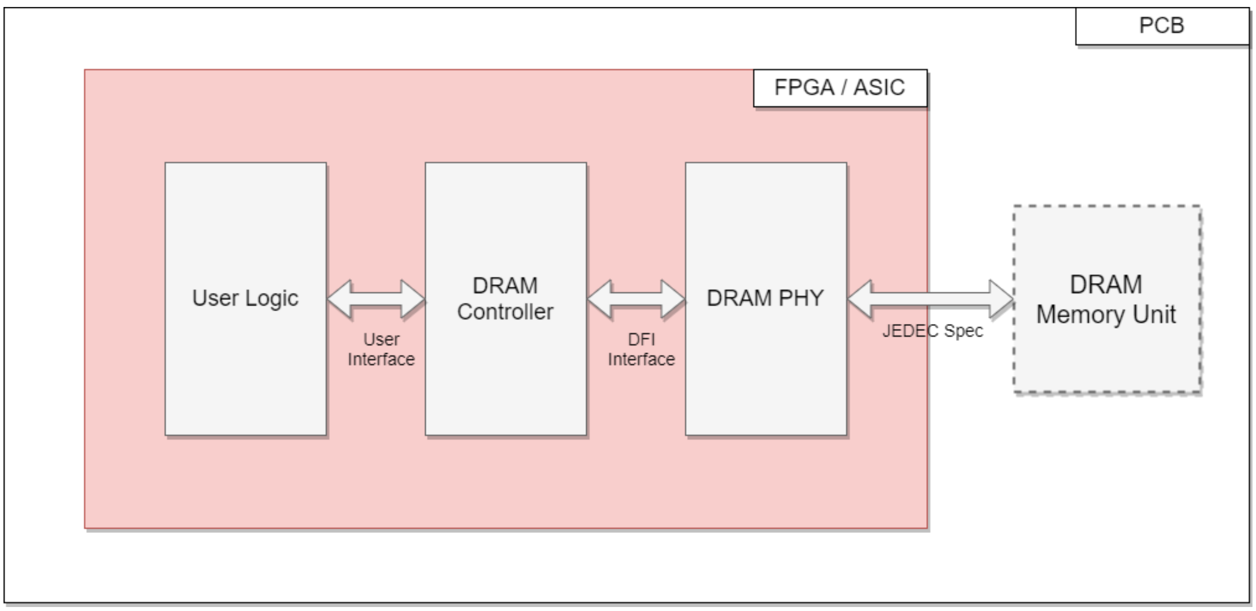

Figure 6: DRAM Subsystem

#tab:DRAMComponents}
Block Description

Physical (PHY) The direct interface to the external DRAM memory Layer bus. Instantiates logic resources to generate the memory clock, control/address signals, and data/data strobes to/from the memory. Executes the DRAM power-up and initialization sequence after system reset. Performs read data capture timing training calibration after system reset, and adjusts read data timing using (IDELAY) elements.

Controller Generates memory commands (Read, Write, Precharge, Refresh) based on commands from the User Interface block. Optionally, can implement a bank management scheme to reduce overhead with opening and closing of bank/rows. The controller logic takes over the DRAM address/control bus after successful completion of DRAM memory initialization and read timing calibration by the PHY layer.

User Interface Custom interface for the user specific application to issue commands and write data to the DRAM memory interface, and to receive read data from the DRAM memory interface.

Clocking / Reset Generates clocks using Digital Clock Manager Logic (DCM) module. Synchronizes resets to the various clock domains used in rest of design.

: DRAM Memory Interface Design Major components & Descriptions

The DRAM is soldered down on the board. The PHY and controller, along with user logic are typically part of the same FPGA or ASIC. The interface between the user logic and the controller can be user defined and need not be standard. When the user logic makes a read or write request to the controller, it issues a logical address. The controller then converts this logical address to a physical address and issues a command to the PHY. The Controller and PHY talk to each other over a standard interface called the DFI interface. The PHY then does all the lower level signaling and drives the physical interface to the DRAM. This interface between the PHY and memory is specified in the JEDEC standard. Think of the controller as the brains and the PHY as the brains.

When you activate a row, the whole page is loaded into the Sense Amplifiers, so multiple reads to an already open page are lesser expensive because you can skip the first step of row activation. The controller typically has the capability to re-order requests issued by the user to take advantage of this. To do the re-ordering it uses a small cache or TCAM and always returns the latest data, so you don't have to worry about stale data or collisions occurring because of this re-ordering done by the controller. The PHY contains the analog drivers and provides the capability to tweak registers to increase drive strength or change terminations, in order to improve signal integrity.

3.2.6 Basic DRAM Controller Operation

DRAM Commands Issued by the Controller: The commands are detected by the memory using these control signals: Row Address Select (RAS), Column Address Select (CAS), and Write Enable (WE) signals. Clock Enable (CKE) is held High after device configuration, and Chip Select (CS) is held Low throughout device operation.

DRAM Memory Commands:

description It is used to deactivate the open row in a particular bank. The bank is available for a subsequent row activation a specified time (tRP) after the Precharge command is issued.

Precharge command:

1. Destructive read: Any read operation that is carried out on a capacitor, will discharge the charges that exist over the capacitor plates leading to a 0 potential layer any reading operation will delete the value stored.
2. The existing value in the row buffer should be stored back for a new read command.
3. The operation of storing the contents in the row buffer back to the appropriate row is known as Precharge.

DRAM devices need to be refreshed regularly after a certain time period. The circuit to flag the Auto Refresh commands is built into the controller. The controller issues an Auto Refresh command after it has completed its current burst. Auto Refresh commands are given the highest priority in the design of the controller.

Before any read or write commands can be issued to a bank within the DRAM memory, a row in the bank must be activated using an active command. After a row is opened, read or write commands can be issued to the row.

The Read command is used to initiate a burst read access to an active row. The values in registers select the bank address & the starting column location in the active row. After the read burst is over, the row is still available for subsequent access until it is precharged.

The Write command is used to initiate a burst write access to an active row. The values in registers select the bank address & the starting column location in the active row. DRAMs use a Write Latency (WL) equal to Read Latency (RL) minus one clock cycle. $ Write Latency = Read Latency - 1 = (Additive Latency + CAS Latency) - 1 $

DRAM Controller Operation is as follows:

- In order to carry out the instruction execution with the help of an instruction pipeline, during the fetch stage cache memory will be used. If the required instruction or data is not available even in the last level cache then DRAM is required.
- Main processing unit has to communicate with the physical DRAM device and that communication is carried out by the DRAM controller. The controller itself has it's own latency called as Controller latency.
- Multiple requests coming from multiple tiles or multiple processors queue up inside the DRAM controller. These requests have to be scheduled in order to be executed by the DRAM controller resulting in Queuing delay & scheduling delay.
- Once scheduling is done, these requests have to be converted into a couple of basic commands.
- Appropriate commands are then propagated from the controller to the physical memory and generating bus latency.
- Once the request reaches the physical memory unit, it has to split into column address and row address.
- There are different scenarios while handling a request:
    - Opened Row Scenario: A scenario where a given row is already kept in the row buffer is known as a open row scenario. In this scenario, only a Column Address Strobe (CAS) is required. There is no need of Activate command.
    - Closed Row Scenario: A scenario where a given row is not already kept in the row buffer is known as a closed row scenario. Access to a closed row is as follows:
        * Activate command
        * Read/Write command
        * Precharge command closes the row and prepares the bank for next access
    - Row conflict Scenario: In this scenario, some other row is already open. In this case row has to be closed first and then Row Address Strobe (RAS) and Column Address Strobe (CAS) has to be given one after another with appropriate timing gap between them.
- The physical memory unit returns the data back to the controller generating bus latency.
- Once data reaches DRAM controller, the controller transfers it to the CPU or the last level cache.

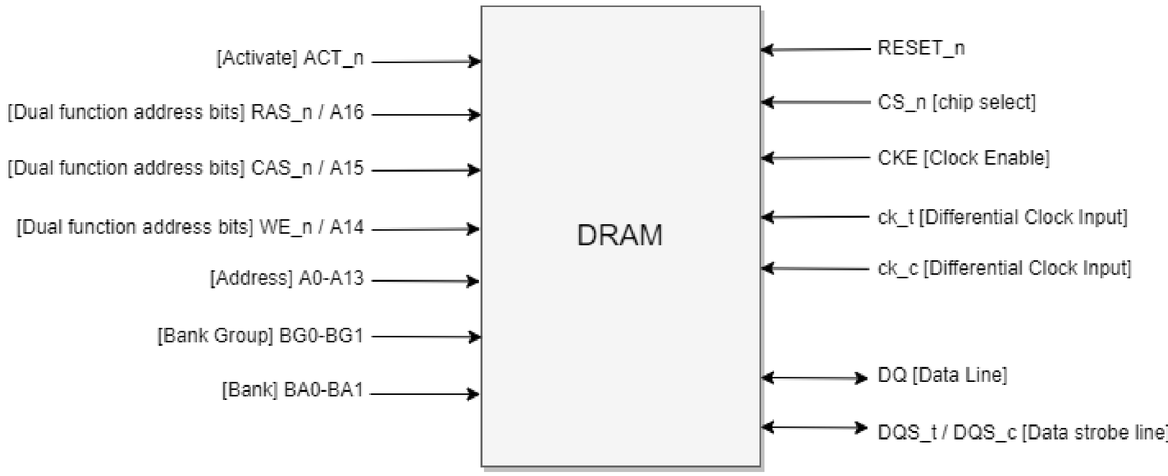

Figure 7: Top Level DRAM block diagram

3.2.7 Internal Physical Structure of DRAM

Usually, DRAM has clock, reset, chip-select, address and data inputs as shown in . The mentions all the pins in detail.

Symbol	Type	Function
RESET_n	Input	DRAM is active only when this signal is HIGH
CS_n	Input	The memory looks at all the other inputs only if this is LOW.
CKE	Input	Clock Enable. HIGH activates internal clock signals and device input buffers and output drivers.
CK_t/CK_c	Input	Differential clock inputs. All address & control signals are sampled at the crossing of posedge of CK_t & negedge of CK_n.
DQ/DQS	Inout	Data Bus & Data Strobe. This is how data is written in and read out. The strobe is essentially a data valid flag.
RAS_n/A16 CAS_n/A15 WE_n/A14	Input	These are dual function inputs. When ACT_n & CS_n are LOW, these are interpreted as Row Address Bits. When ACT_n is HIGH, these are interpreted as command pins to indicate READ, WRITE or other commands.
ACT_n	Input	Activate command input
BG0-1 BA0-1	Input	Bank Group, Bank Address
A0-13	Input	Address inputs

Figure 8: DRAM block diagram

3.3 DDR RAM

In case of the next generation of the SDR DRAM (Single Data Rate DRAM or synchronous DRAM), the data is transferred twice during the clock cycle. First, during the positive edge and secondly, during the negative edge of the clock cycle. And that is why this generation of the SDRAM is known as the **Double Data Rate** or **DDR SDRAM**.

There are different generations of DDR RAM ranging from the DDR1 up to the DDR4 which is considered to be the latest. The first generation of DDR RAM is known as the DDR1 RAM. As compared to the SDR SDRAM, the voltage levels has been reduced from 3.3V to the 2.5V.

There are a total two types of frequencies associated with DRAM:

Input output clock frequency is the frequency at which the data is being transferred between the RAM and the memory controller.

RAM Internal clock frequency of the RAM is the frequency which is being used by the RAM for the internal operations.

In case of SDRAM, input output clock frequency and the internal clock frequency of the RAM are same. For PC-100 specification on the SDRAM module means that the input output clock frequency is 100 mega transfers per second and if the data bus is 64 bit wide, then the data rate in terms of the bits per second will be 100 MHz into 64 bits. Which is 800 Megabytes per second.

In case of DDR RAM, the data is being transferred both during the rising as well as the falling edge of the clock cycle. Hence, in a single clock cycle, instead of a single bit, 2 bits are pre-fetched which is known as the **2 bit pre-fetch**. In case of DDR1 RAM, the internal clock frequency, as well as the input output bus clock frequency, are same. Generally, DDR1 RAM is operated in the range of 133 MHz up to 200 MHz. But if you see the data rate at the input output bus, it will be double compared to the clock frequency. In case of DDR1 RAM, the data is transferred both during rising as well as the falling edge. For suppose if DDR1 RAM is operated at 133 MHz then the data rate will be 266 Mega transfer per second. If the bus frequency is 200 MHz then the data transfer rate will be 400 Mega transfer per second. And if the input output bus is 64 bits wide, then the data rate will be 3200 Megabytes per second.

Nowadays, DDR RAMs are generally denoted by the term DDR followed by the transfer rate of this RAM. For a DDR1 module or a DDR1 stick, most probably will have a specification like PC-3200. It means that the maximum speed or the maximum bandwidth which can be achieved by this DDR1 RAM is 3200 Megabytes per second.

3.4 DDR2 RAM

After the first generation, the second generation of DDR RAM is DDR2 RAM. DDR2 SDRAM superseded the original DDR1 SDRAM specification, and was superseded by DDR3 SDRAM when launched in 2007. The maximum capacity on commercially available DDR2 DIMMs is 8GB, but chipset support and availability for those DIMMs is sparse and more common 2GB per DIMM are used. In case of DDR2 RAM, it is operated at 1.8 V instead of 2.5 V unlike the DDR1 RAM. The internal RAM clock frequency is same as the previous generation. Instead the data rate is doubled compared to the first generation which was achieved by increasing the number of bits that are being pre-fetched during each cycle. In case of this DDR2 RAM instead of 2 bits, 4 bits are pre-fetched during each cycle. In other words, the internal bus width of DDR2 RAM has been doubled when compared with DDR1.

For suppose if the input output bus is 64 bits wide, then the internal bus width of this RAM will be equal to 128 bits. So, in this way, in a single cycle, this RAM can handle double amount of data. To handle the same amount of data, the clock frequency of this input output bus should be get doubled. Suppose DDR2 RAM, is operated at 100 MHz internal clock frequency then the input output bus should have the clock frequency of 200 MHz. And in case of this DDR RAM, as data is transferred both during rising and falling edge, so the data rate will be doubled compared to the clock frequency, that is 400 mega transfer per second. Suppose if DDR2 RAM is operated at 400

MHz clock frequency, then the data rate will be equal to 800 mega transfer per second. And in terms of DDR terminology, it can be written as DDR2-800 or PC2-6400.

3.5 DDR3 RAM

After the second generation, the third generation of DDR RAM is the DDR3 RAM. DDR3 SDRAM superseded the original DDR2 SDRAM specification, and was superseded by DDR4 SDRAM when launched in 2014. The DDR3 standard permits DRAM chip capacities of up to 8 gibibits (Gibit), and up to four ranks of 64 bits each for a total maximum of 16 gibibytes (GiB) per DDR3 DIMM. In case of this DDR3 RAM, the voltage is further reduced from 1.8V to the 1.5V. The internal clock frequency of DDR3 RAM is slightly improved compared to DDR2. But the data rate that you can achieve with the same frequency has been doubled as compared to the DDR2. In case of DDR3 RAM, the number of bits that is being pre-fetched has been further increased from 4 bits to the 8 bits. In other words, the internal data bus width of RAM has been increased 2 times compared to DDR2 and 4 times to DDR1.

For suppose if the internal clock frequency is 100 MHz, then to match the data rate, the input output bus should be get operated at the 4 times the clock frequency that is 400 MHz. And the transfer rate will be 800 mega transfer per second. For DDR3-800 followed by PC3-6400 on any DDR3 RAM, means that the clock frequency of this RAM is 400 MHz and the maximum transfer rate which can be achieved is 800 mega transfer per second. Maximum bandwidth of the RAM is 6400 Megabytes per second.

3.6 DDR4 RAM

DDR4 SDRAM is the abbreviation for 'Double Data Rate fourth generation synchronous dynamic random-access memory', the latest variant of memory in computing. DDR4 is able to achieve higher speed and efficiency thanks to increased transfer rates and decreased voltage. The primary advantages of DDR4 over its predecessor, DDR3, include higher module density and lower voltage requirements, coupled with higher data rate transfer speeds. The DDR4 standard allows for DIMMs of up to 64 GiB in capacity, compared to DDR3's maximum of 16 GiB per DIMM.

After the third generation, the fourth generation of DDR RAM is DDR4 RAM. DDR4 SDRAM superseded the original DDR3 SDRAM specification, and was superseded by DDR5 SDRAM when launched in 2020. The DDR4 standard allows for DIMMs of up to 64 GiB in capacity, compared to DDR3's maximum of 16 GiB per DIMM. In case of this DDR4 RAM, the operating voltage has been further reduced from 1.5V to the 1.2V. The number of bits that are being pre-fetched is same as DDR3 i.e. 8 bits per cycle. In case of this DDR4 RAM, the internal clock frequency of the RAM has been increased. For suppose if you are operating at 400 MHz then the clock frequency of the input output bus should be 4 times, that means 1600 MHz. The transfer rate will be equal to 3200 Mega transfer per second. Module terminology DDR4-3200 followed by PC4-25600. 25600 is the speed in terms of Megabytes per second.

- **Single channel mode:** In single channel mode, the physical RAM, uses the usual input output bus width (64 bits).
- **Dual channel mode** In dual channel mode, the same physical RAM, uses twice the input output bus width to effectively achieve twice the data rate. Suppose, there is an 8 GB DDR4 RAM running in single channel mode and two 4 GB DDR4 RAMs running in dual channel

mode, then the bandwidth that can be achieved with two 4 GB of DDR4 RAM will be better as compared to the single channel 8GB of DDR4 RAM.

3.7 Application specific DDR versions

A compact version of DIMM module is known as Small Outline DIMM Modules (SO-DIMM). Another version of dynamic RAMs which are used inside the mobile or smartphones are known as the mobile DDR or Low Power DDR. Low Power DDR RAMs are also having different generations. Starting from LPDDR1 up to the LPDDR4. LPDDR RAMs are optimised for the low power consumption. Another specially catered version of DDR RAMs which is used for graphics cards is known as the graphics DDR or GDDR. As this Graphical DDR is used for the multimedia applications, the data handling is quite extensive. Hence, GDDR RAMs have larger bandwidth compared to usual DDRs.

3.8 DDR Packaging

The older generations of DRAMs were available in the Dual Inline Package(DIP). Then after the next generation of RAMs were available in the Single In-Line Modules (SIMM). In Single In-Line module, the memory chips are soldered onto the one PCB, and the pins are available on the single side of the PCB. And that is a reason, it is known as the Single In-Line Modules. Single In-line module can provide data bus width of 32 bits. But suppose if you want 64 bits of the data bus, then you need to connect the two single In-line modules in the parallel.

After the next generation of RAMs were available in the Dual In-Line Module or DIMM. In Dual In-Line Module, it is possible to have 64 bits wide data bus. Also, the pins are available both in front as well as the back of the PCB. And that is a reason, it is known as the Dual In-Line Module.

All the DDR generations have a different number of pins as well as the different operating voltage. Hence, all the four generation of RAMs are not either forward or backward compatible. So, suppose a motherboard supporting DDR3 RAM, will not support either DDR2 or DDR4 RAM.

3.9 Xilinx DDR MIG Controller IP

The Memory Interface Generator (MIG) generates DDR4 SDRAM, DDR3 SDRAM, DDRII SRAM, DDR SDRAM, DDR2 SDRAM, QDRII SRAM, and RLDRAM II interfaces for various Xilinx FPGAs. The tool takes inputs such as the memory interface type, FPGA family, FPGA devices, frequencies, data width, memory mode register values, and so forth, from the user through a graphical user interface (GUI). The tool generates RTL, SDC, UCF, and document files as output. RTL or EDIF (EDIF is created after running a script file, where the script file is a tool output) files can be integrated with other design files.

MIG is a tool used to generate memory interfaces for Xilinx FPGAs. MIG generates Verilog or VHDL RTL design files, user constraints files (UCF), and script files. The script files are used to run simulations, synthesis, map, and par for the selected configuration.

4 Flash Memory

Flash memory is an electronic (solidstate) nonvolatile computer memory storage medium that can be electrically erased and reprogrammed. The two main types of flash memory are named after the NAND and NOR logic gates. The individual flash memory cells, consisting of floatinggate

MOSFETs (floatinggate metaloxidesemiconductor fieldeffect transistors), exhibit internal characteristics similar to those of the corresponding gates. While EPROMs had to be completely erased before being rewritten, NANDtype flash memory may be erased, written and read in blocks (or pages) which are generally much smaller than the entire device. NORtype flash allows a single machine word (byte) to be written to an erased location or read independently. A flash memory device typically consists of one or more flash memory chips (each holding many flash memory cells) along with a separate flash memory controller chip. The NAND type is found primarily in memory cards, USB flash drives, solidstate drives (those produced in 2009 or later), and similar products, for general storage and transfer of data. NAND or NOR flash memory is also often used to store configuration data in numerous digital products, a task previously made possible by EEPROM or batterypowered static RAM.

Serial Flash Serial flash is a small, lowpower flash memory that provides only serial access to the data rather than addressing individual bytes, the user reads or writes large contiguous groups of bytes in the address space serially. Serial Peripheral Interface Bus (SPI) is a typical protocol for accessing the device. When incorporated into an embedded system, serial flash requires fewer wires on the PCB than parallel flash memories, since it transmits and receives data one bit at a time. This may permit a reduction in board space, power consumption, and total system cost. A flash memory controller (or flash controller) manages data stored on flash memory and communicates with a computer or electronic device. Flash memory controllers can be designed for operating in low dutycycle environments like SD cards, Compact Flash cards, or other similar media.

4.1 QSPI Flash

AXI Quad SPI LogiCORE IP AXI Quad Serial Peripheral Interface (SPI) core connects the AXI4 interface to those SPI slave devices that support the Standard, Dual, or Quad SPI protocol instruction set. This core provides a serial interface to SPI slave devices. The Dual/Quad SPI is an enhancement to the standard SPI protocol (described in the Motorola M68HC11 data sheet) and provides a simple method for data exchange between a master and a slave.

Configurable SPI modes:

- Standard SPI mode
- Dual SPI mode
- Quad SPI mode
- Programmable SPI clock

5 EMMC

eMMC stands for embedded MultiMedia Card and refers to a package consisting of both flash memory and a flash memory controller. The controller here is divided into 2 parts: 1. Host controller 2. Device controller. Host controller sits on the host device and is usually operates on a higher layer of programming. Device controller sits inside the flash memory hardware. It takes signals controlled by the Host controller and converts them into interpretable signals for the flash memory. The eMMC specification covers the behavior of the interface and the device controller. As part of this specification the existence of a host controller and a memory storage array are implied but the operation of these pieces is not fully specified.

5.1 eMMC Device Overview

The eMMC device transfers data via a configurable number of data bus signals. The communication signals are:

- **CLK** Each cycle of this signal directs a one bit transfer on the command and either a one bit (1x) or a two bits transfer (2x) on all the data lines. The frequency may vary between zero and the maximum clock frequency.
- **Data Strobe** This signal is generated by the device and used for data output and CRC status response output in HS400 mode. The frequency of this signal follows the frequency of CLK. For data output each cycle of this signal directs two bits transfer(2x) on the data one bit for positive edge and the other bit for negative edge. For CRC status response output, the CRC status is latched on the positive edge only, and don't care on the negative edge.
- **CMD** This signal is a bidirectional command channel used for device initialization and transfer of commands. The CMD signal has two operation modes: opendrain for initialization mode, and pushpull for fast command transfer. Commands are sent from the eMMC host controller to the eMMC device and responses are sent from the device to the host.
- **DAT0DAT7** These are bidirectional data channels. The DAT signals operate in pushpull mode. Only the device or the host is driving these signals at a time. By default, after power up or reset, only DAT0 is used for data transfer. A wider data bus can be configured for data transfer, using either DAT0DAT3 or DAT0DAT7, by the eMMC host controller. The eMMC device includes internal pullups for data lines DAT1DAT7. Immediately after entering the 4bit mode, the device disconnects the internal pull ups of lines DAT1, DAT2, and DAT3. Correspondingly, immediately after entering to the 8bit mode the device disconnects the internal pullups of lines DAT1DAT7.

All communication between host and device are controlled by the host (master). The host sends a command, which results in a device response. Five operation modes are defined for the eMMC system (hosts and devices):

- Boot mode The device will be in boot mode after power cycle, reception of CMD0 with argument of 0xF0F0F0F0 or the assertion of hardware reset signal.
- Device identification mode The device will be in device identification mode after boot operation mode is finished or if host and /or device does not support boot operation mode. The device will be in this mode, until the SET_RCA command (CMD3) is received.
- Interrupt mode Host and device enter and exit interrupt mode simultaneously. In interrupt mode there is no data transfer. The only message allowed is an interrupt service request from the device or the host.
- Data transfer mode The device will enter data transfer mode once an RCA is assigned to it. The host will enter data transfer mode after identifying the device on the bus.
- Inactive mode The device will enter inactive mode if either the device operating voltage range or access mode is not valid. The device can also enter inactive mode with GO_INACTIVE_STATE command (CMD15). The device will reset to Pre-idle state with power cycle.

If the CMD line is held LOW for 74 clock cycles and more after powerup or reset operation (either

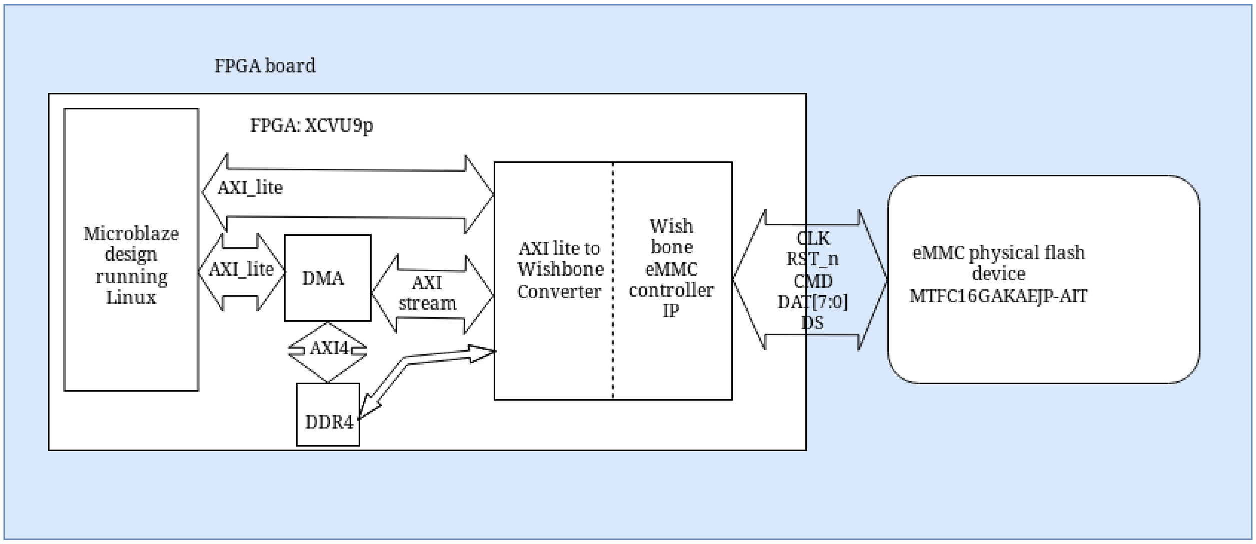

Figure 9: Internal block diagram of EMMC IP

through CMD0 with the argument of 0xF0F0F0F0 or assertion of hardware reset for eMMC, if it is enabled in Extended CSD register byte [162], bits [1:0]) before the first command is issued, the slave recognizes that boot mode is being initiated and starts preparing boot data internally. Timing diagram of EMMC IP Boot up sequence is shown in

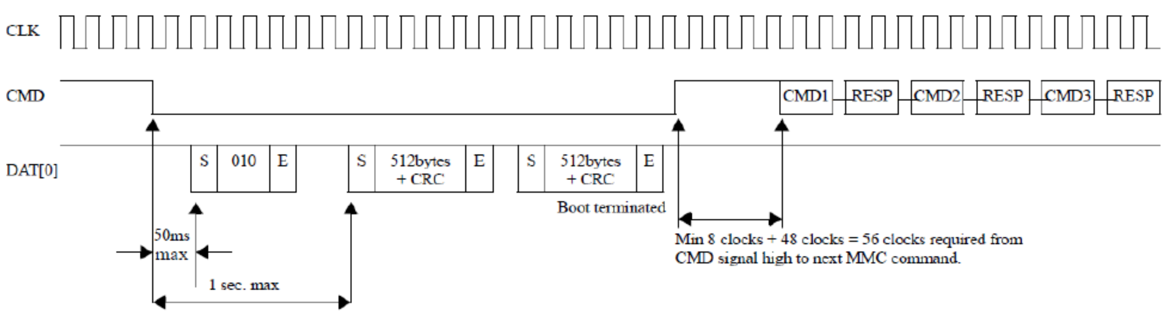

Figure 10: Timing diagram of EMMC IP Boot up sequence

The partition from which the master will read the boot data can be selected in advance using EXT_CSD byte [179], bits [5:3]. The data size that the master can read during boot operation can be calculated as 128KB X BOOT_SIZE_MULT (EXT_CSD byte [226]). Within 1 second after the CMD line goes LOW, the slave starts to send the first boot data to the master on the DAT line(s). The master must keep the CMD line LOW to read all of the boot data. The master must use push-pull mode until boot operation is terminated.

The master can choose to use single data rate mode with backward-compatible interface timing, single data rate with high-speed interface timing or dual data rate timing (if it supported) shown in 10.6 by setting a proper value in EXT_CSD register byte [177] bits [4:3]. EXT_CSD register byte [228], bit 2 tells the master if the high-speed timing during boot is supported by the device. The master can also choose to use the dual data rate mode with interface during boot by setting '10' in EXT_CSD register byte [177], bits [4:3]. EXT_CSD register byte [228], bit 1 tells the master if the dual data rate mode during boot is supported by the device.

The master can choose to receive boot acknowledge from the slave by setting '1' in EXT_CSD register, byte [179], bit 6, so that the master can recognize that the slave is operating in boot mode. If boot acknowledge is enabled, the slave has to send acknowledge pattern '010' to the master within 50ms after the CMD line goes LOW. If boot acknowledge is disabled, the slave will not send out acknowledge pattern '0-1-0.' In the single data rate mode, data is clocked out by the device and sampled by the host with the rising edge of the clock and there is a single CRC per data line.

In the dual data rate mode, data is clocked out with both the rising edge of the clock and the falling edge of the clock and there are two CRC appended per data line. In this mode, the block length is always 512 bytes, and bytes come interleaved in either 4-bit or 8-bit width configuration. Bytes with odd number (1,3,5, ... ,511) shall be sampled on the rising edge of the clock by the host and bytes with even number (2,4,6, ... ,512) shall be sampled on the falling edge of the clock by the host. The device will append two CRC16 per each valid data line, one corresponding to the bits of the 256 odd bytes to be sampled on the rising edge of the clock by the host and the second for the remaining bits of the 256 even bytes of the block to be sampled on the falling edge of the clock by the host. All timings on DAT lines shall follow DDR timing mode. The start bit, the end bit and Boot acknowledge bits are only valid on the rising edge of the clock. The value of the falling edge is not guaranteed. The master can terminate boot mode with the CMD line HIGH. If the master pulls the CMD line HIGH in the middle of data transfer, the slave has to terminate the data transfer or acknowledge pattern within NST clock cycles (one data cycle and end bit cycle). If the master terminates boot mode between consecutive blocks, the slave must release the data line(s) within NST clock cycles.

Boot operation will be terminated when all contents of the enabled boot data are sent to the master. After boot operation is executed, the slave shall be ready for CMD1 operation and the master needs to start a normal MMC initialization sequence by sending CMD1. Please find the boot sequence in which will be operated by the eMMC driver in the hindsight for initialization and then for the operation of the eMMC memory.

6 Gigabit Ethernet

1G/10G Ethernet IP contains PCS, PMA, Phy management and reset controller.

Transceiver: Combination tx/rx used when sending high-speed digital data/control signals acreoss physical Medium. Used in PHY layer of OSI model. Made up of the physical coding sublayer (PCS) and physical medium attachment (PMA).

PCS: Digital logic that prepares and formats data for TX across a physical medium type or restores RX data to original form. Ex. Encoding, decoding, scrambling, descrambling.

PMA: Converts digital data to serial analog streams or reverse.

Data link Layer Concerned with packaging data into frames and transmitting those frames on the network, performing error detection/correction and uniquely identifying network devices with an address(MAC) and flow control

MAC (Media Access Control):

- Physical addressing: 48 bit address assigned to a device's network interface card (NIC)

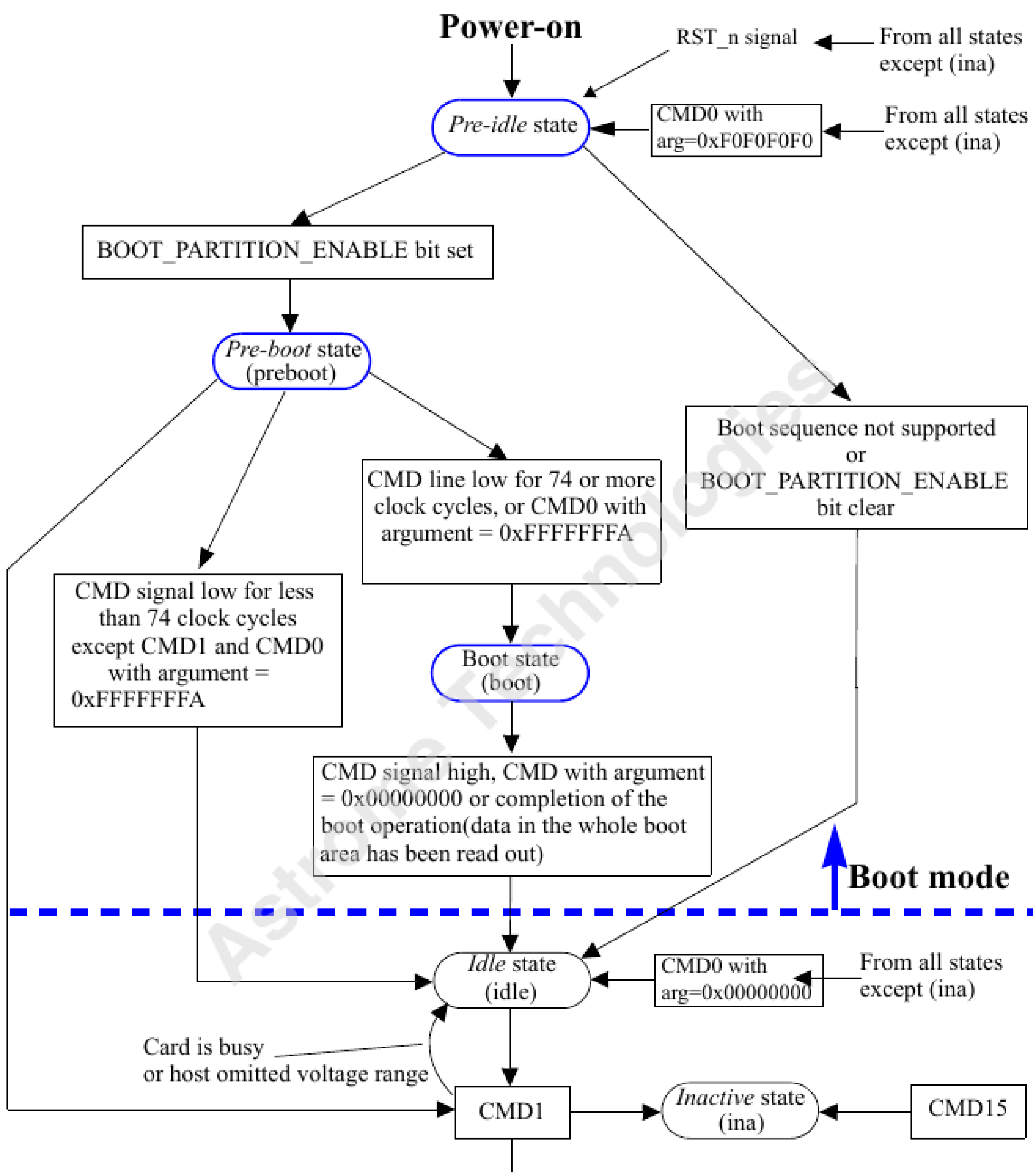

Figure 11: EMMC IP Boot up sequence

- Logical topology: Logical network topologies
- Method of transmitting: CSMA/CD

LLC (Link Layer Control):

- Connection services: provides for acknowledgement of receipt of a message.
  - Flow control: Limits amount of data sender can send at one time
  - Error Control: Allows rx to let tx know when an expected data frame wasn't received or was corrupted by using a checksum.
- Synchronizing transmissions

6.1 1G/2.5G Ethernet IP

In computer networking, Gigabit Ethernet (GbE or 1 GigE) is the term applied to transmitting Ethernet frames at a rate of a gigabit per second (1 billion bits per second) and is defined by the IEEE 802.3ab standard. There are five physical layer standards for Gigabit Ethernet using optical fiber (1000BASEX), twisted pair cable (1000BASET), or shielded balanced copper cable (1000BASECX). The IEEE 802.3z standard includes 1000BASESX for transmission over multimode fiber, 1000BASELX for transmission over singlemode fiber, and the nearly obsolete 1000BASECX for transmission over shielded balanced copper cabling. These standards use 8b/10b encoding, which inflates the line rate by 25%, from 1000 Mbit/s to 1250 Mbit/s, to ensure a DC balanced signal. The symbols are then sent using NRZ. Optical fiber transceivers are most often implemented as userswappable modules in SFP form or GBIC on older devices. IEEE 802.3ab, which defines the widely used 1000BASET interface type, uses a different encoding scheme in order to keep the symbol rate as low as possible, allowing transmission over twisted pair.

The AXI Ethernet Subsystem provides a control interface to internal registers via a 32bit AXI4Lite Interface subset. This AXI4Lite slave interface supports single beat read and write data transfers (no burst transfers). The transmit and receive data interface is via the AXI4Stream interface. This core has been designed incorporating the applicable features described in IEEE Std. 802.3. This core supports the use of MII, GMII, SGMII, RGMII, and 1000BASEX interfaces to connect a media access control (MAC) to a physicalside interface (PHY) chip. The internals of 1G/2.5G Ethernet Subsystem IP is shown in .

The subsystem provides an AXI4Lite bus interface for a simple connection to the processor core to allow access to the registers. This AXI4Lite slave interface supports single beat read and write data transfers (no burst transfers). 32bit AXI4Stream buses are provided for moving transmit and receive Ethernet data to and from the subsystem. These buses are designed to be used with an AXI Direct Memory Access (DMA) IP core, AXI4Stream Data FIFO, or any other custom logic in any supported device. The AXI4Stream buses are designed to provide support for TCP/UDP partial or full checksum offload in hardware if required. The PHY side of the subsystem is connected to an offtheshelf Ethernet PHY device, which performs the BASET standard at 1 Gb/s, 100 Mb/s, and 10 Mb/s speeds. The PHY device can be connected using any of the following supported interfaces: GMII/MII, RGMII, or, by using the 1G/2.5G Ethernet PCS/PMA or SGMII module.

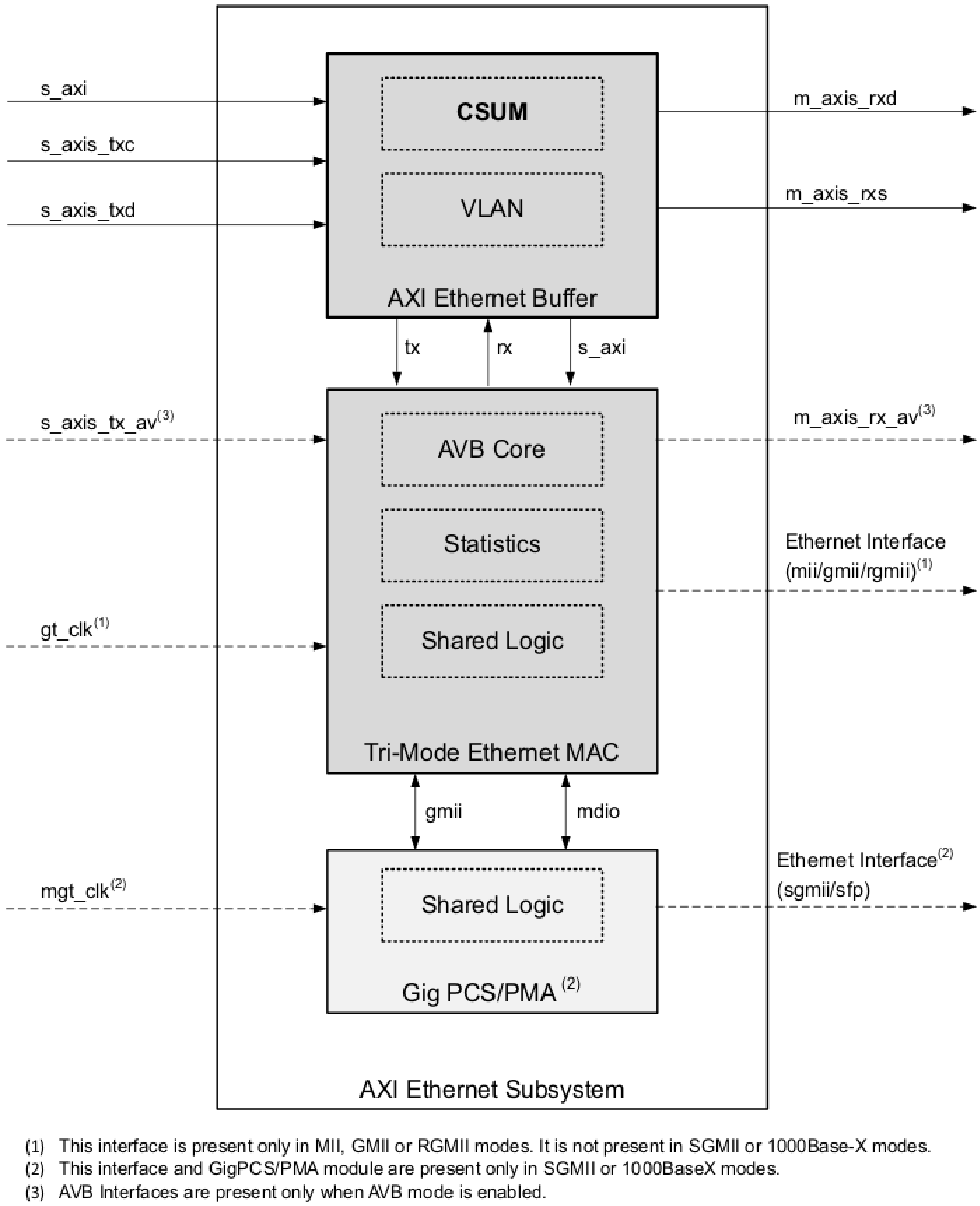

Figure 12: Internal block diagram of 1G/2.5G Ethernet Subsystem IP

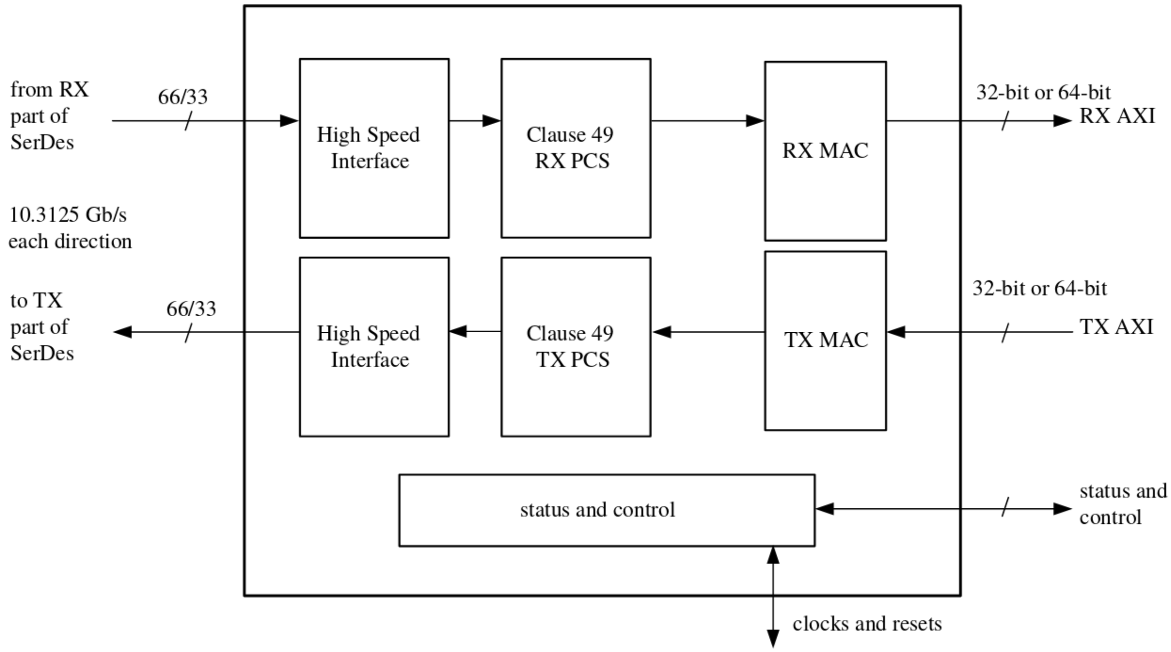

Figure 13: Internal block diagram of 10G/25G Ethernet Subsystem IP

6.2 10G/25G Ethernet IP

The Xilinx LogiCORE IP 10G/25G Ethernet solution provides a 10 Gigabit or 25 Gigabit per second (Gbps) Ethernet Media Access Controller integrated with a PCS/PMA in BASER/KR modes or a standalone PCS/PMA in BASER/KR modes. The core is designed to work with the latest Xilinx UltraScale and UltraScale+ FPGAs. The 25G Ethernet IP is designed to the new 25 Gb/s Ethernet Consortium standard and supports the demand of cloud data centers to enable lower cost and increased performance solutions between the server and the top of rack switch and to increase the front panel density by two. The internals of 10G/25G Ethernet Subsystem IP is shown in .

7 Direct Memory Access